

Frontal theta oscillations during emotion regulation in people with borderline personality disorder

Основная информация

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Аннотация

<jats:sec id="S2056472424000176_sec_a1"> Background Borderline personality disorder (BPD) is a severe psychiatric disorder conceptualised as a disorder of emotion regulation. Emotion regulation has been linked to a frontolimbic network comprising the dorsolateral prefrontal cortex and the amygdala, which apparently synchronises its activity via oscillatory coupling in the theta frequency range. <jats:sec> <jats:sec id="S2056472424000176_sec_a2"> Aims To analyse whether there are distinct differences in theta oscillatory coupling in frontal brain regions between individuals with BPD and matched controls during emotion regulation by cognitive reappraisal. <jats:sec> <jats:sec id="S2056472424000176_sec_a3" sec-type="methods"> Method Electroencephalogram (EEG) recordings were performed in 25 women diagnosed with BPD and 25 matched controls during a cognitive reappraisal task in which participants were instructed to downregulate negative emotions evoked by aversive visual stimuli. Between- and within-group time-frequency analyses were conducted to analyse regulation-associated theta activity (3.5–8.5 Hz). <jats:sec> <jats:sec id="S2056472424000176_sec_a4" sec-type="results"> Results Oscillatory theta activity differed between the participants with BPD and matched controls during cognitive reappraisal. Regulation-associated theta increases were lower in frontal regions in the BPD cohort compared with matched controls. Functional connectivity analysis for regulation-associated changes in the theta frequency band revealed a lower multivariate interaction measure (MIM) increase in frontal brain regions in persons with BPD compared with matched controls. <jats:sec> <jats:sec id="S2056472424000176_sec_a5" sec-type="conclusions"> Conclusions Our findings support the notion of alterations in a frontal theta network in BPD, which may be underlying core symptoms of the disorder such as deficits in emotion regulation. The results add to the growing body of evidence for altered oscillatory brain dynamics in psychiatric populations, which might be investigated as individualised treatment targets using non-invasive stimulation methods. <jats:sec>

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